UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Artificial Intelligence

TOPIC 10:
SUPERVISED LEARNING MODELLING AND EVALUATION: CLASSIFICATION USING SUPPORT VECTOR MACHINES

MACHINE LEARNING (SAIA 2113)

Session 2025/2026 - Sem 1

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Topic 9: Supervised Learning Modelling And Evaluation: Classification Using Support Vector Machine

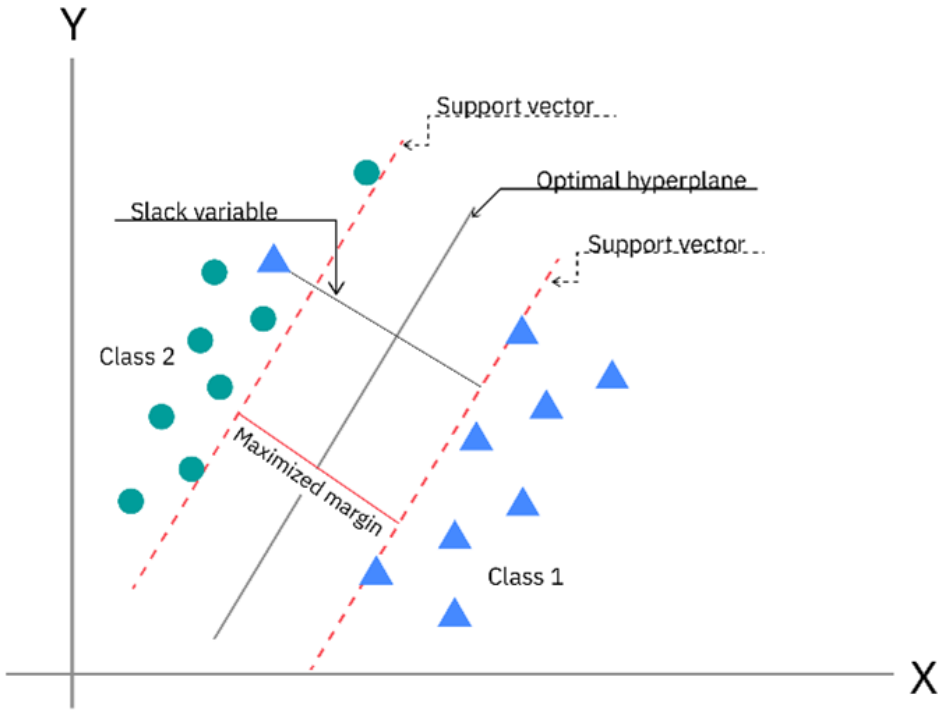
- Introduction to SVM
- Terminology
- Application
- Types
- How does SVMs work?
- Advantages
- Disadvantages

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Introduction

- A support vector machine (SVM) is a supervised machine learning algorithm.
- Developed in the 1990s by Vladimir N. Vapnik and his colleagues.
- They published this work in a paper titled "Support Vector Method for Function Approximation, Regression Estimation, and Signal Processing“ in 1995.
- SVMs are commonly used within classification problems, regression and outliers’ detection.
- Classifies data by finding an optimal line or hyperplane that maximizes the distance between each class in an N-dimensional space.

Introduction



- The lines that are adjacent to the optimal hyperplane are known as support vectors as these vectors run through the data points that determine the maximal margin.
- In this approach, each data point is represented as a point in an n-dimensional space, where n is the number of features.
- The goal is to find a hyperplane that best separates the two classes, which is used to minimize an error.
- The core idea of SVM is to find a maximum marginal hyperplane (MMH) that best divides the dataset into classes.
- This finds the best line (or hyperplane) to separate data into groups, maximizing the distance between the closest points (support vectors) of each group.

Introduction

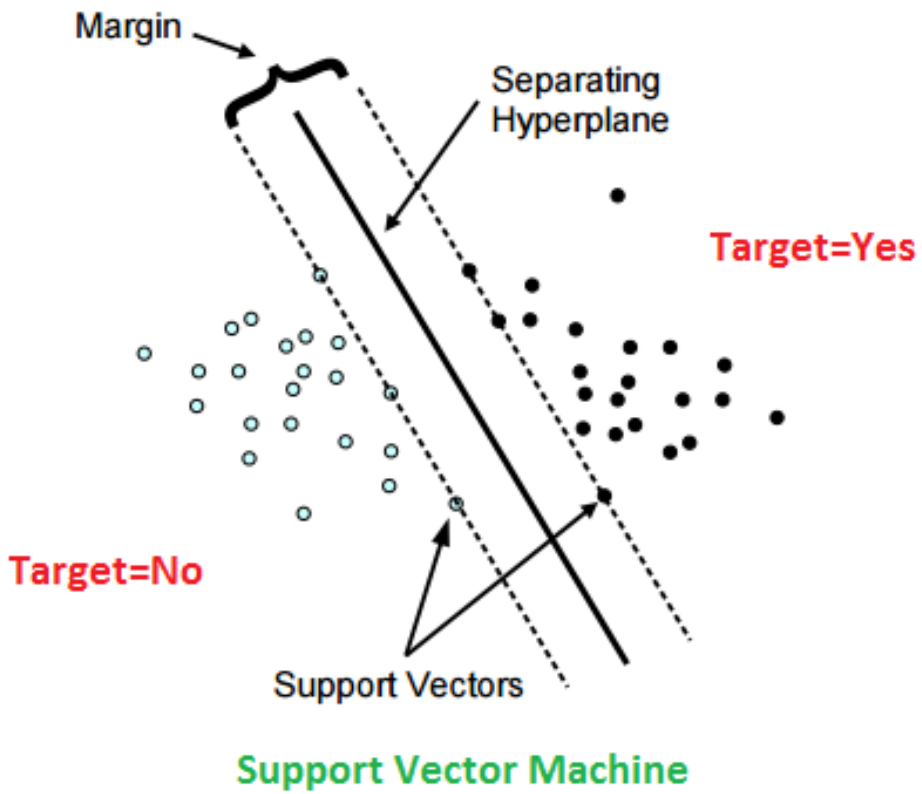
- The SVM algorithm can handle both linear and nonlinear classification tasks.
- However, when the data is not linearly separable (complex data), kernel functions are used to transform the data higher-dimensional space to enable linear separation.
- This application of kernel functions can be known as the “kernel trick”, and the choice of kernel function depends on data characteristics and the specific use case.
- Known as discriminative classifier (separates data points using a hyperplane with the largest amount of margin).
- Easily handle multiple continuous and categorical variables.

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Terminology

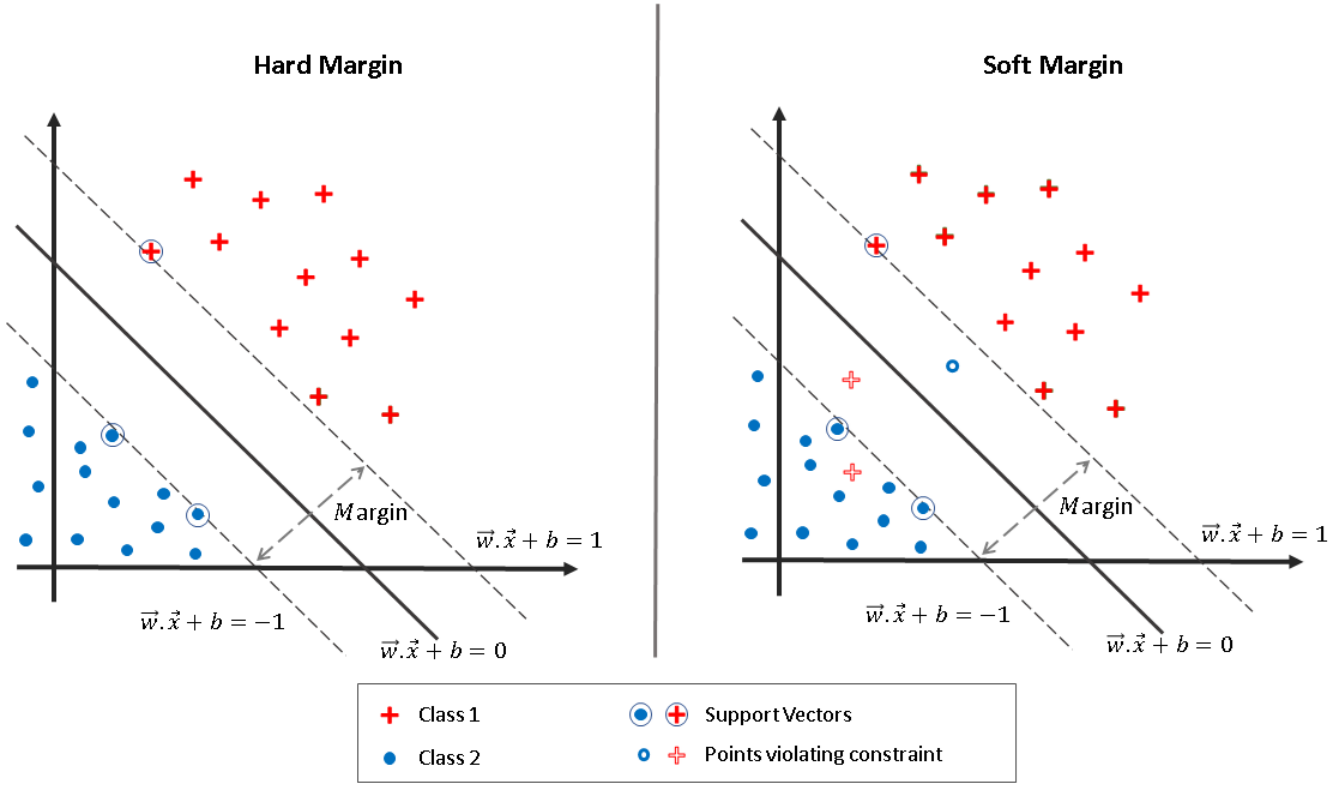


- **Hyperplane**: A hyperplane is a decision boundary which separates between given set of data points having different class labels. The SVM classifier separates data points using a hyperplane with the maximum amount of margin, represented by the equation $wx + b = 0$ in linear classification.

- **Support Vectors**: The closest data points to the hyperplane, crucial for determining the hyperplane and margin in SVM.

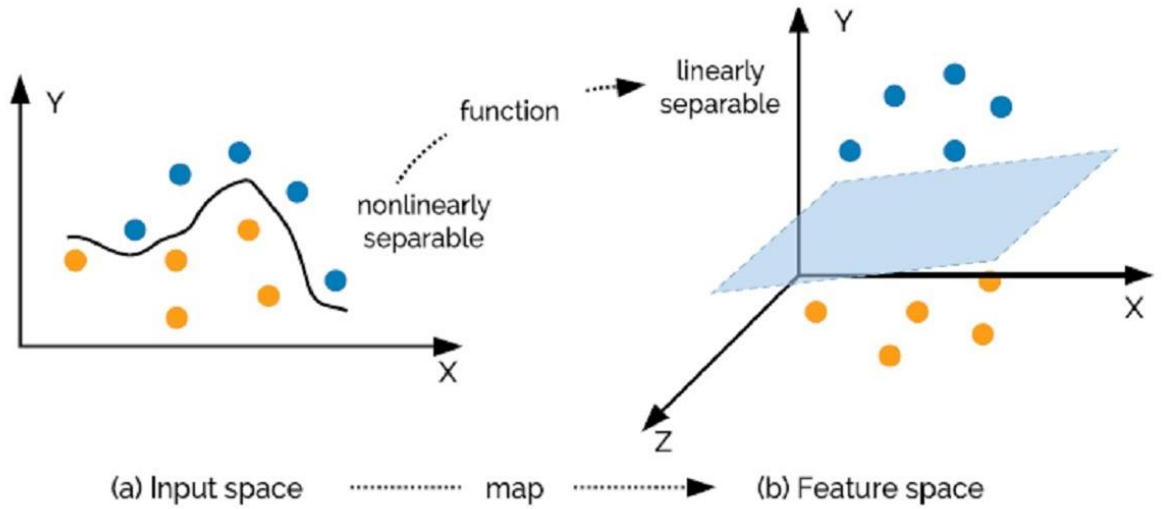
- **Margin**: The distance between the hyperplane and the support vectors. SVM aims to maximize this margin for better classification performance.

Terminology



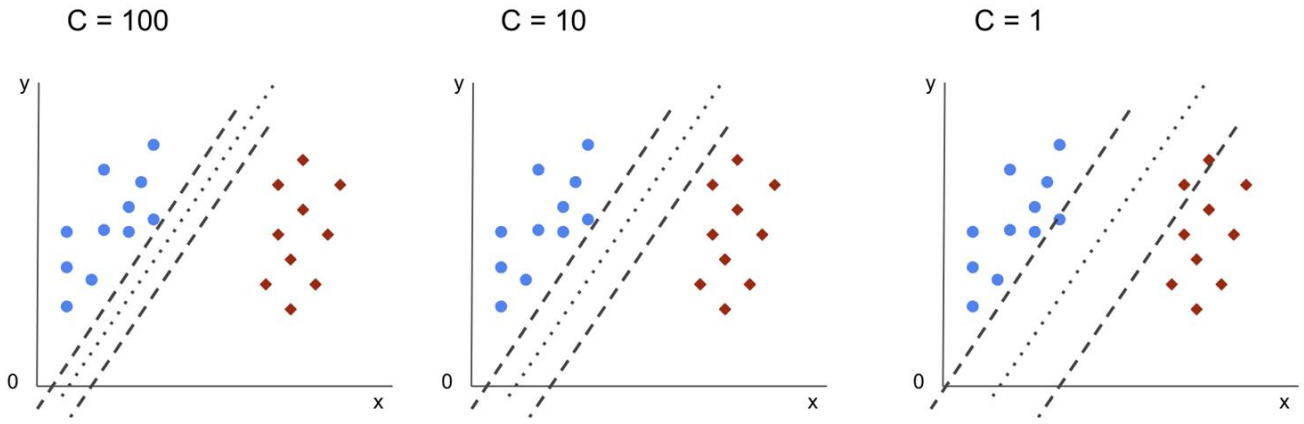
- **Hard Margin**: A maximum-margin hyperplane that perfectly separates the data without misclassifications.
- **Soft Margin**: Allows some misclassifications by introducing slack variables, balancing margin maximization and misclassification penalties when data is not perfectly separable.

Terminology



- **Kernel**: transformation of input space to higher dimensional space, enabling SVM to handle non-linearly separable data.

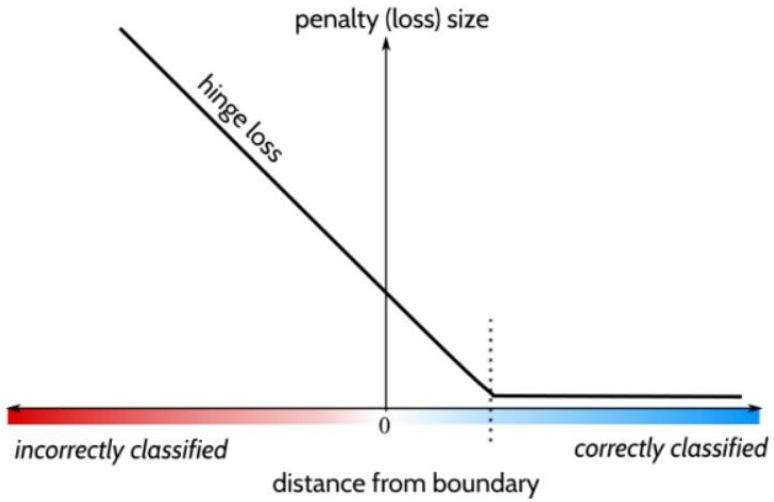
Terminology



- **C**: A regularization parameter balancing margin maximization and misclassification penalties. A higher C value enforces a stricter penalty for misclassifications.
- The parameter C controls the trade-off between having a wide margin and minimizing classification errors.

- Left: C = 100 (High penalty for misclassification): Aim to classify all points correctly, even if that leads to a narrower margin. Tight margin, closely hugging support vectors. Risk: Overfitting the data.
- Middle: C = 10 (Moderate penalty): Balance between margin size and classification accuracy. Margin is wider than for C = 100, and might allow some misclassification or close calls. Good generalization in many cases.
- Right: C = 1 (Low penalty for misclassification): The model allows more slack (soft margin) to tolerate some misclassifications for a wider margin. Better for noisy data or when avoiding overfitting is crucial. Risk: Underfitting the data.

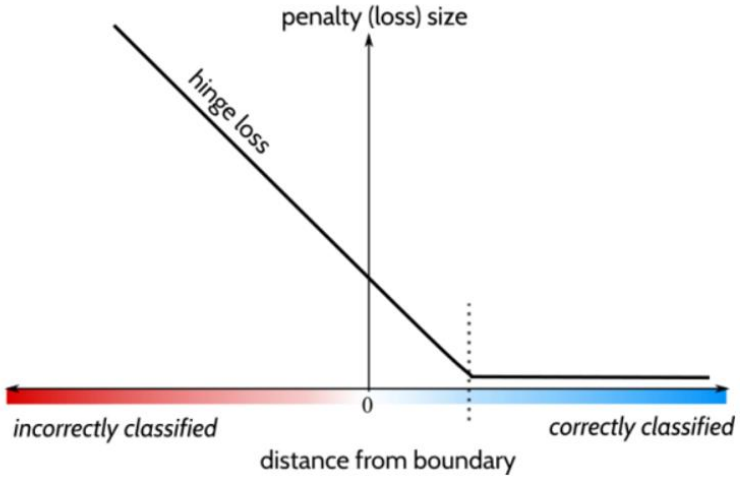
Terminology



- **Hinge Loss**: A loss function penalizing misclassified points or margin violations, combined with regularization in SVM.

- x-axis: Distance from the boundary of any single instance.
- y-axis: loss size, or penalty, that the function will incur depending on its distance.

Terminology



- That dotted line on the x-axis represents the number 1. This means that when an instance's distance from the boundary is greater than or at 1, our loss size is 0.
- If the distance from the boundary is 0 (meaning that the instance is literally on the boundary), then we incur a loss size of 1.
- We see that correctly classified points will have a small(or none) loss size, while incorrectly classified instances will have a high loss size.
- A negative distance from the boundary incurs a high hinge loss. This essentially means that we are on the wrong side of the boundary, and that the instance will be classified incorrectly.
- On the flip size, a positive distance from the boundary incurs a low hinge loss, or no hinge loss at all, and the further we are away from the boundary(and on the right side of it), the lower our hinge loss will be.

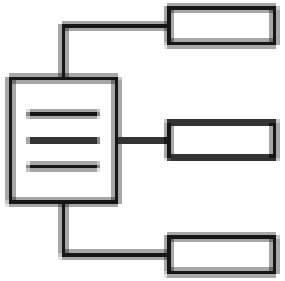
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Application of SVM Classification

Text Classification



Natural language processing (NLP) for tasks such as sentiment analysis, spam detection, and topic modeling.

Image Classification



- Object detection and image retrieval.
- Useful in security domains, classifying an image as one that has been tampered with.

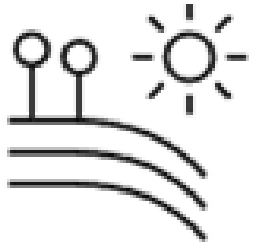
Application of SVM Classification

Bioinformatics



- Protein classification, gene expression analysis, and disease diagnosis.
- Applied in cancer research because they can detect subtle trends in complex datasets.

Geographic Information System (GIS)



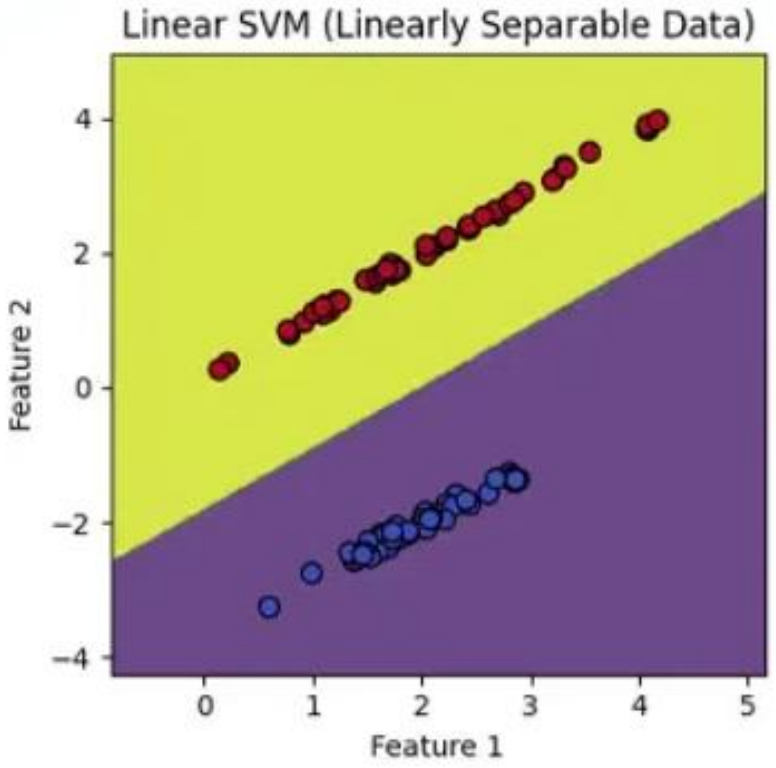
- Analyse layered geophysical structures underground, filtering out the 'noise' from electromagnetic data.
- Predict the seismic liquefaction potential of soil, which is relevant to field of civil engineering.

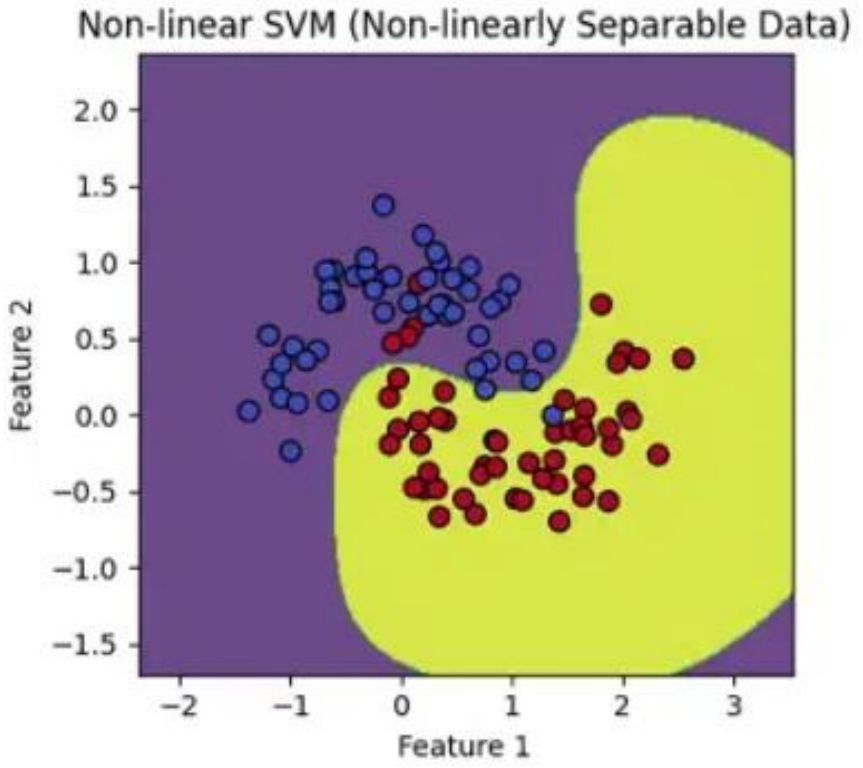
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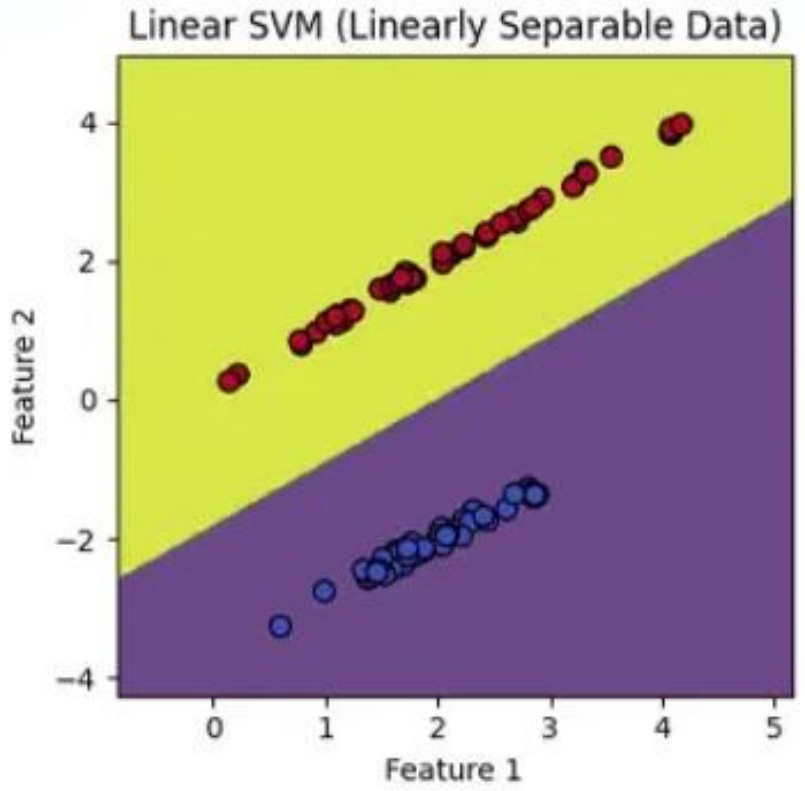
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Types of SVM Model



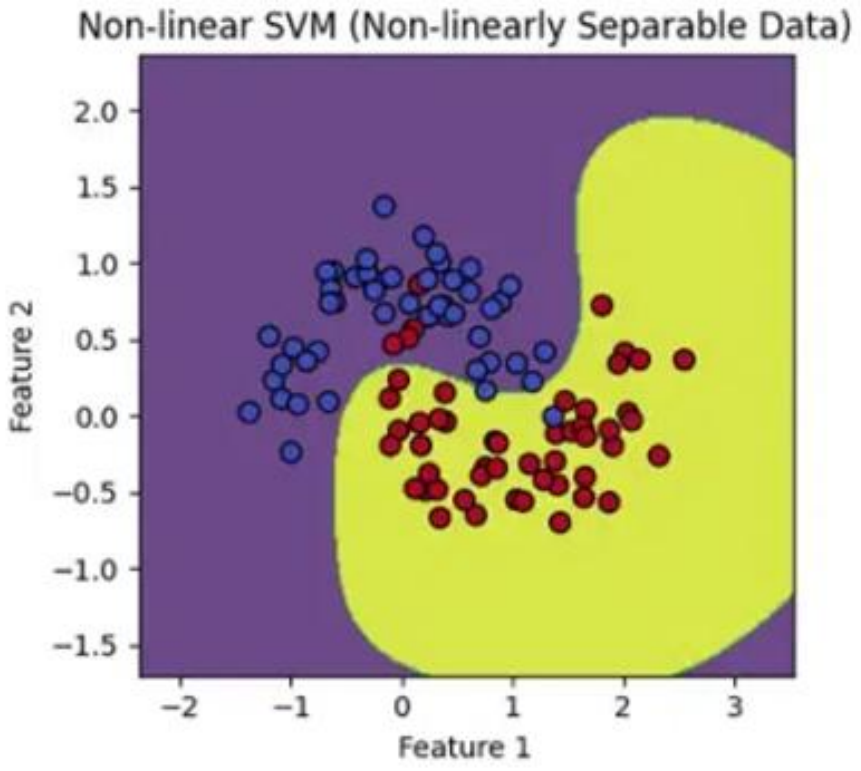


Types of SVM Model – Linear SVM



- When the data is perfectly linearly separable only then we can use Linear SVM.
- $wx + b = 0$
- where w is the weight vector, x is the input vector, and b is the bias term.
- Linear SVMs use a linear decision boundary to separate the data points of different classes.
- The data points can be classified into their respective classes by using a single straight line (in 2D) or a hyperplane (in higher dimensions)
- A hyperplane that maximizes the margin between the classes is the decision boundary.

Types of SVM Model – Non-Linear SVM



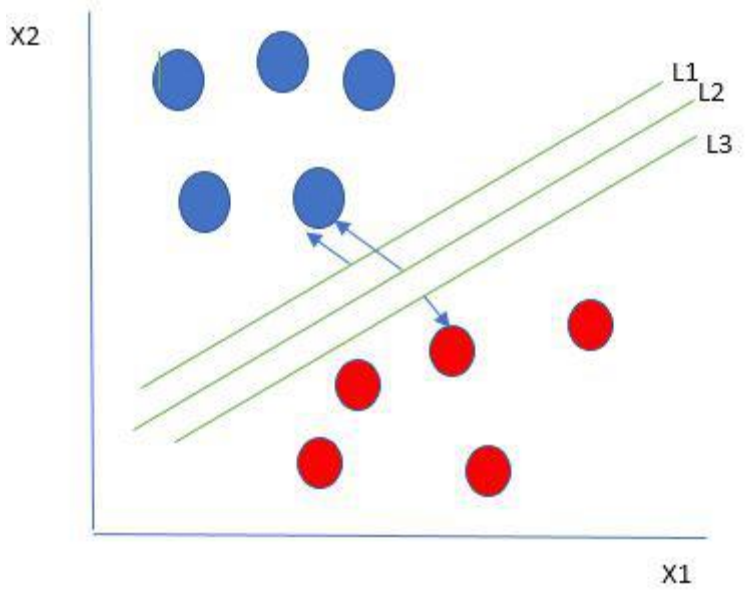
- Non-Linear SVM can be used to classify data when it cannot be separated into two classes by a straight line (in the case of 2D).
- By using kernel functions, nonlinear SVMs can handle nonlinearly separable data.
- The original input data is transformed by these kernel functions (called as kernel tricks) into a higher-dimensional feature space, where the data points can be linearly separated.
- A linear SVM is used to locate a nonlinear decision boundary in this modified space.

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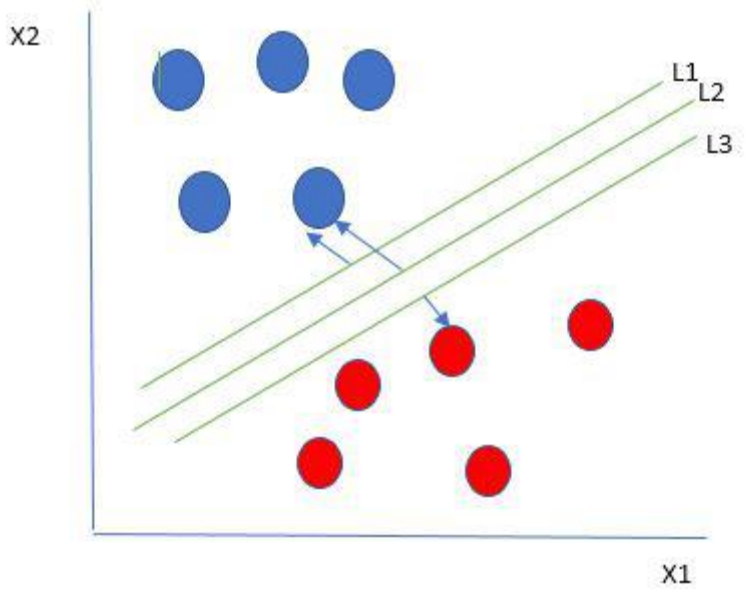
How Does SVMs Work?



Multiple hyperplanes separate the data from two classes

- The key idea behind the SVM algorithm is to **find the decision boundary, straight line (in 2D graph) and hyperplane (in more dimensions)** that best separates two classes by maximizing the margin between them.
- This margin is the distance from the hyperplane to the nearest data points (support vectors) on each side.
- The best hyperplane, also known as the "**hard margin**," is the one that maximizes the distance between the hyperplane and the nearest data points from both classes.
- This ensures a clear separation between the classes. So, from the above figure, we choose L2 as hard margin.

How Does SVMs Work?



Multiple hyperplanes separate the data from two classes

- Consider a binary classification problem with two classes, labeled as +1 and -1. We have a training dataset consisting of input feature vectors X and their corresponding class labels Y.
- The equation for the linear hyperplane can be written as:

$$w \cdot x + b = 0$$

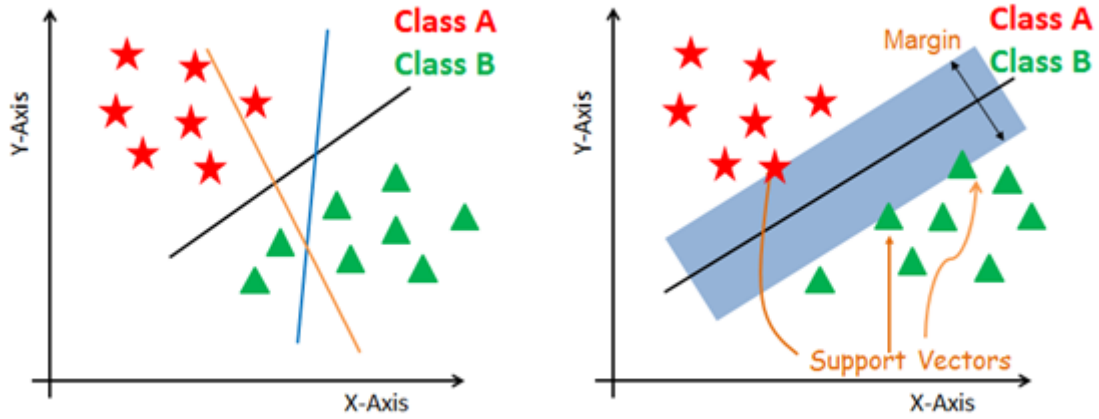
Where:

w is the normal vector to the hyperplane (the direction perpendicular to it).

b is the offset or bias term, representing the distance of the hyperplane from the origin along the normal vector w.

x is the point on feature space.

How Does SVMs Work?



SVM searches for the maximum marginal hyperplane in the following steps:

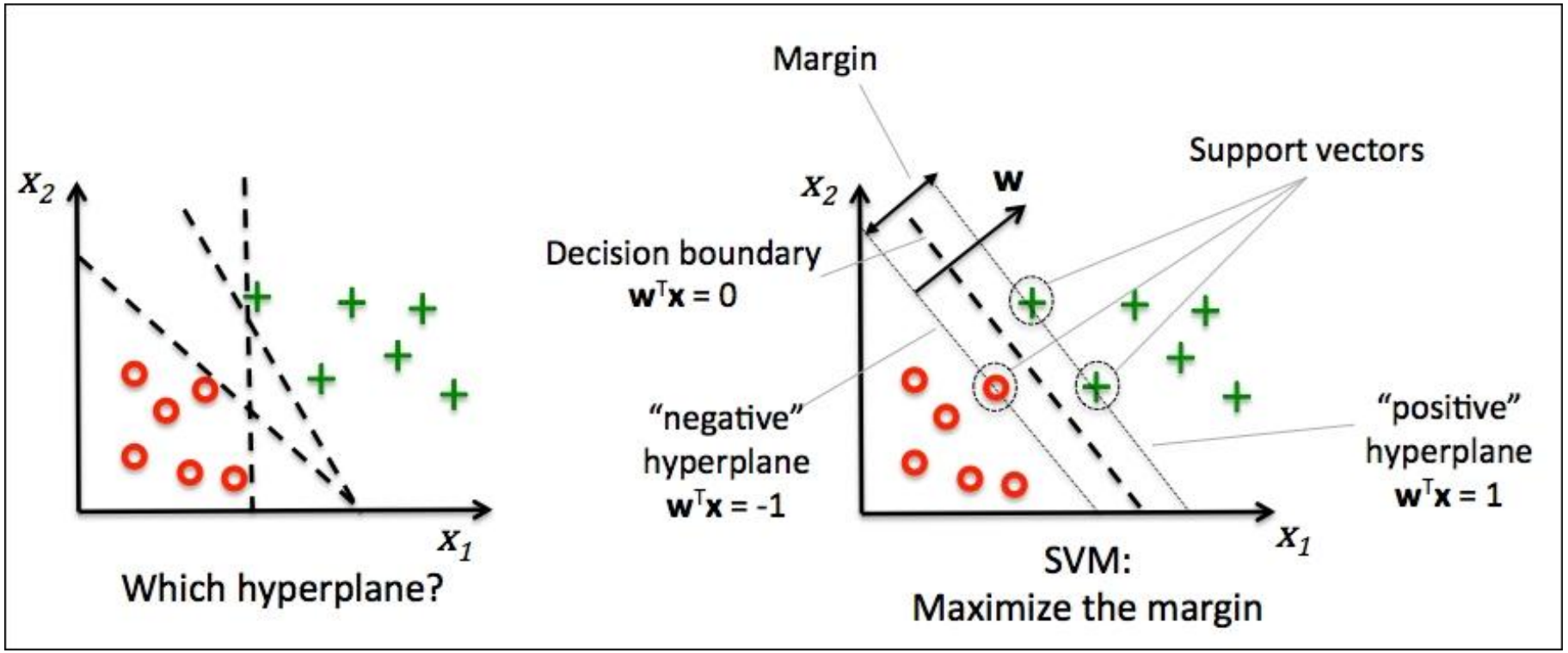
1. Generate hyperplanes which segregates the classes in the best way. Left-hand side figure showing **three hyperplanes black, blue and orange**. Here, the **blue and orange have higher classification error**, but the **black is separating the two classes correctly**.

1. The **best hyperplane** is that plane that **has the maximum distance from both the classes (maximum margin)** and choose the one which is **farthest from the data points**. It is known as the **maximum margin hyperplane** and the linear classifier it defines is known as a **maximum margin classifier**.

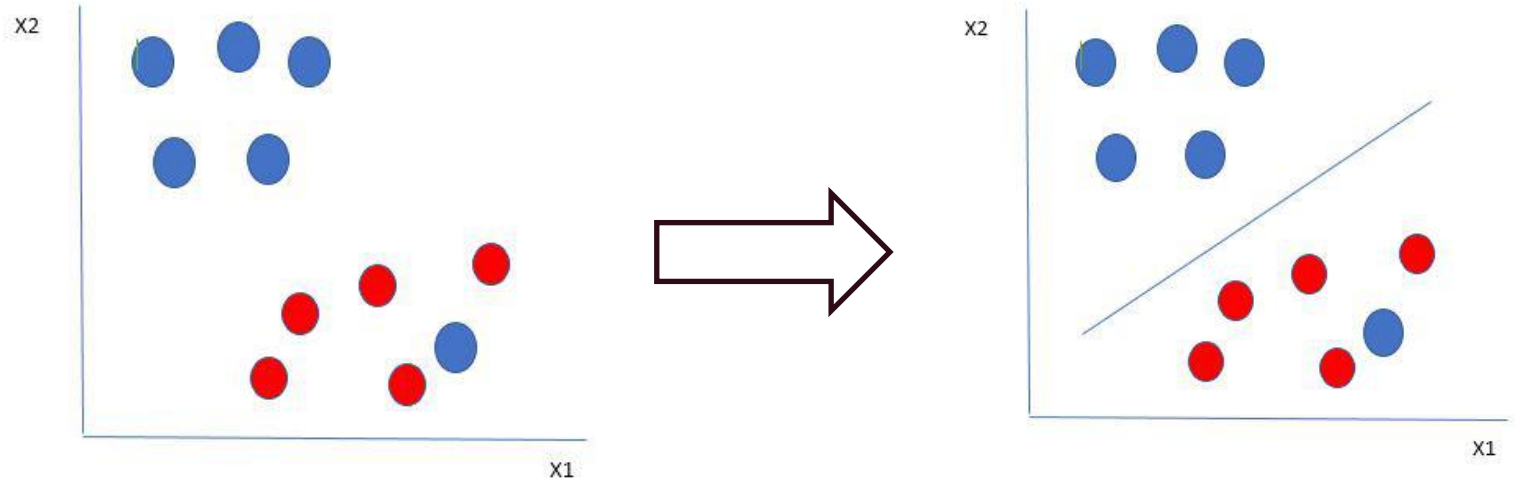
How Does SVMs Work?

The following diagram illustrates the concept of maximum margin and maximum margin hyperplane in a clear manner.

Maximum margin hyperplane



How Does SVMs Work?



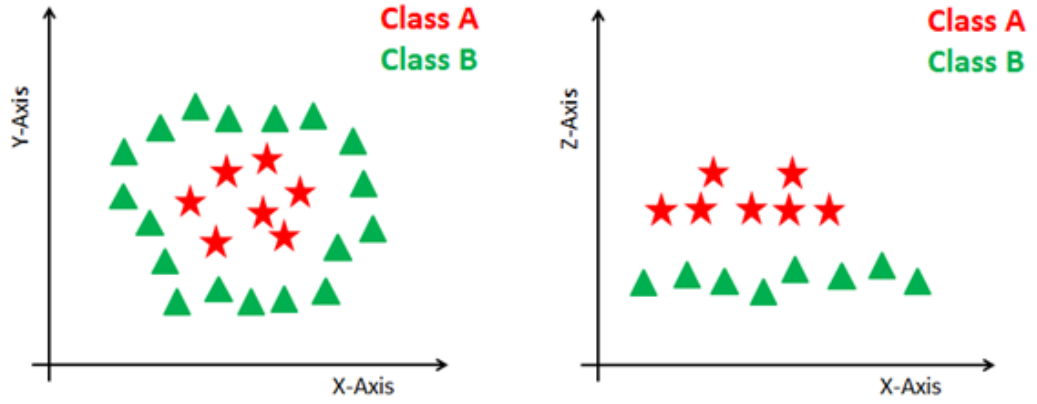
Selecting hyperplane for data with outlier

- The **blue ball** in the boundary of red ones is an **outlier** of blue balls.
- The SVM algorithm has the **characteristics to ignore the outlier** and finds the best hyperplane that maximizes the margin. SVM is robust to outliers.
- A soft margin allows for some misclassifications or violations of the margin to improve generalization.
- The SVM optimizes the following equation to balance margin maximization and penalty minimization:

$$\text{Objective Function} = \left(\frac{1}{\text{margin}}\right) + \lambda \sum \text{penalty}$$

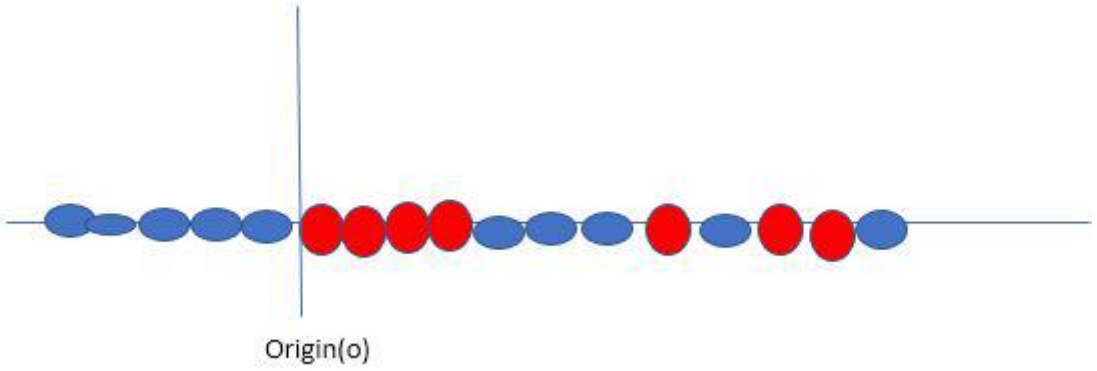
- The penalty used for violations is often **hinge loss**.

How Does SVMs Work?

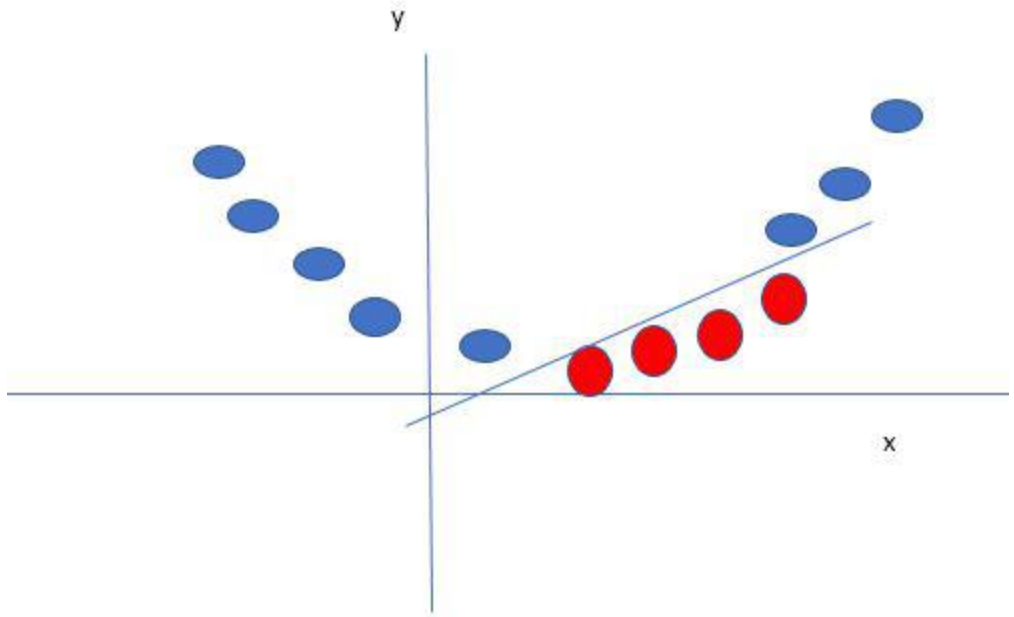


- **Some problems can't be solved using linear hyperplane**, as shown in the figure above (left-hand side).
- In such situation, SVM uses a **kernel trick** to transform the input space to a higher dimensional space as shown on the right figure.
- The data points are plotted on the x-axis and z-axis (Z is the squared sum of both x and y: $z = x^2 = y^2$). Now you can easily segregate these points using linear separation.

How Does SVMs Work?



Original 1D dataset for classification



Mapping 1D data to 2D to become able to separate the two classes

In this case, the new variable y is created as a function of distance from the origin.

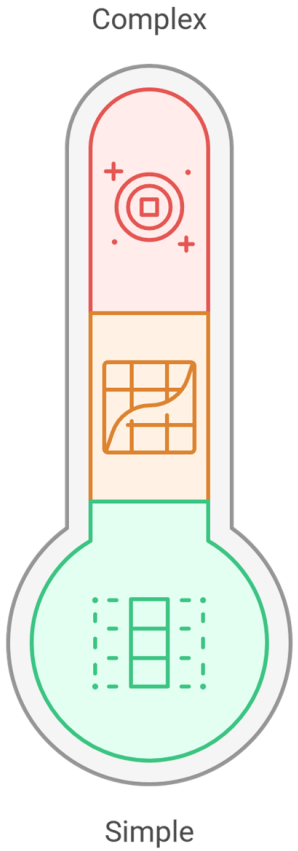
How Does SVMs Work?

- The “kernel trick” helps to **reduce** some of that **complexity**, making the **computation** more **efficient**, and it does this by replacing dot product calculations with an equivalent kernel function.

- The kernel takes a **low-dimensional input space (2D) and transforms it into a higher dimensional space (3D)**. In other words, you can say that it converts non separable problem to separable problems by adding more dimension to it. It is most useful in non-linear separation problem.

- A kernel is a **function that maps data points into a higher-dimensional space without explicitly computing the coordinates in that space**.

How Does SVMs Work? - Kernel



RBF Kernel

Transforms data using distance relationships.

Polynomial Kernel

Maps data into polynomial feature space.

Linear Kernel

Suited for linearly separable data.

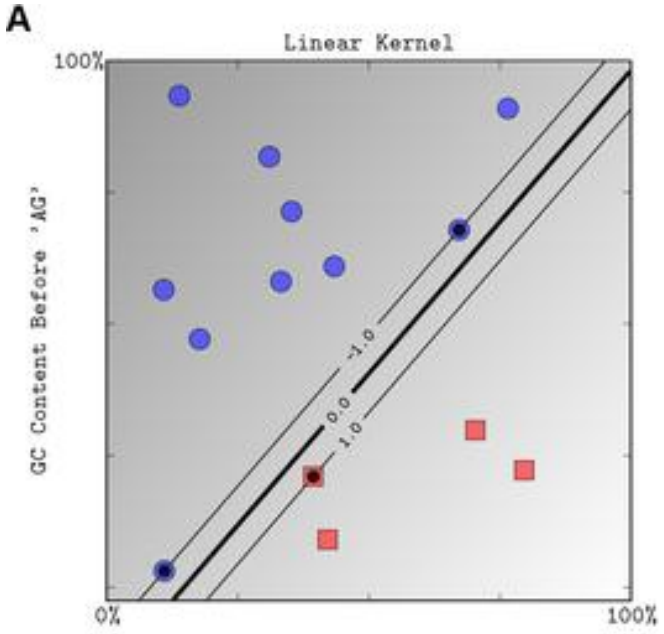
How Does SVMs Work?

1. **Linear Kernel**:

- Used as normal dot product any two given observations.
- The product between two vectors is the sum of the multiplication of each pair of input values.

$$K(x, xi) = sum(x * xi)$$

- Used when the data is linearly separable. It means that data can be separated using a single line.
- It is mostly used when there are large number of features in a dataset.
- Linear kernel is often used for text classification purposes.
- Training with a linear kernel is usually faster, because we only need to optimize the C regularization parameter.
- When training with other kernels, we also need to optimize the γ (gamma) parameter. So, performing a grid search will usually take more time.



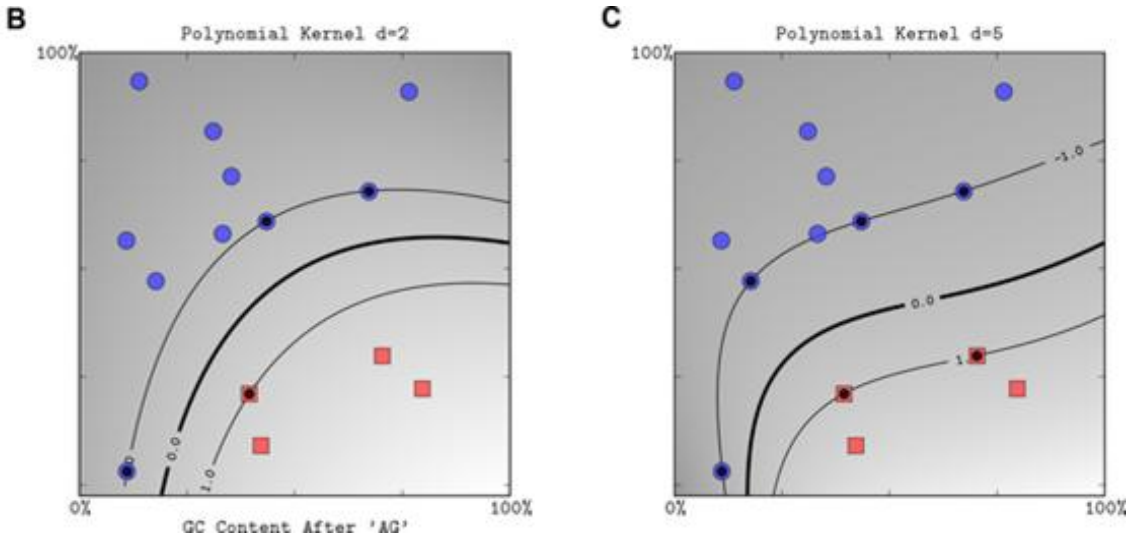
How Does SVMs Work?

2. **Polynomial Kernel**:

- More generalized form of the linear kernel.
- The polynomial kernel can distinguish curved or nonlinear input space.

$$K(x, xi) = 1 + sum(x * xi)^d$$

- Where d is the degree of the polynomial.
- $d = 1$ is similar to the linear transformation.
- The degree needs to be manually specified in the learning algorithm.
- Represents the similarity of vectors (training samples) in a feature space over polynomials of the original variables.
- Not only at the given features of input samples to determine their similarity, but also combinations of the input samples.
- Polynomial kernel is very popular in Natural Language Processing.
- The most common degree is $d = 2$ (quadratic), since larger degrees tend to overfit on NLP problems.



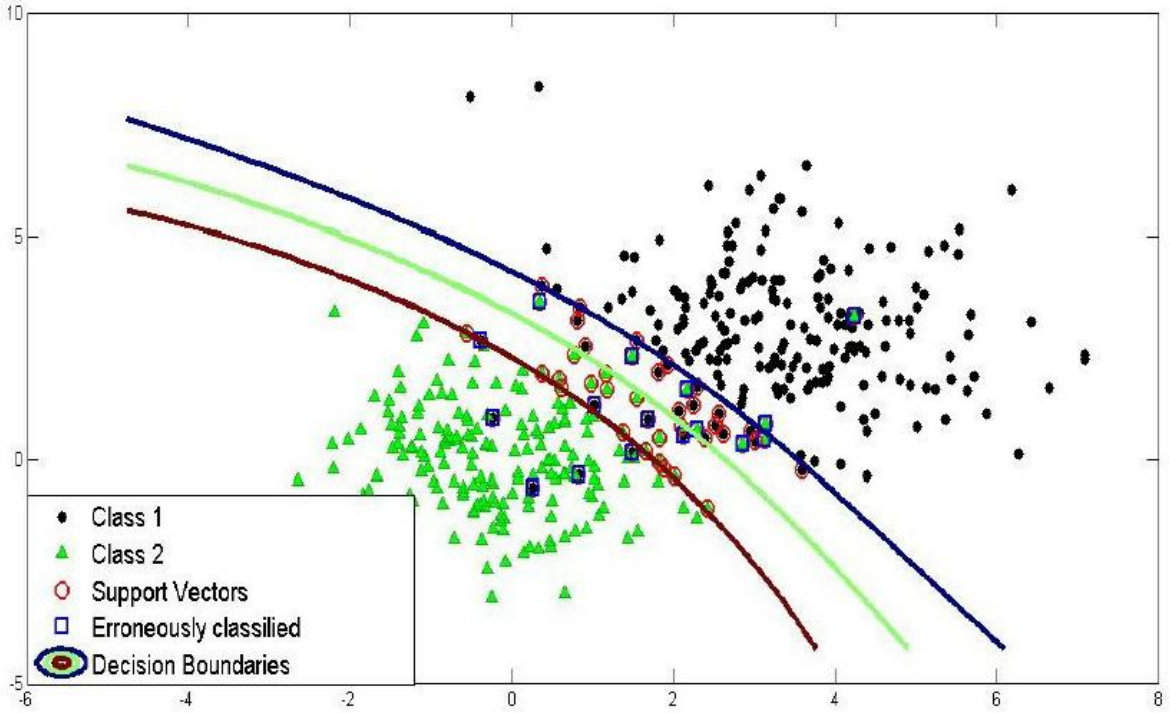
How Does SVMs Work?

3. **Radial Basis Function (RBF) Kernel** :

- Map an input space in infinite dimensional space.

$$K(x, xi) = \exp(-gamma * sum((x - xi\text{^}2))$$

- Here gamma is a parameter, which ranges from 0 to 1.
- A higher value of gamma will perfectly fit the training dataset, which causes over-fitting.
- Gamma=0.1 is considered to be a good default value.
- The value of gamma needs to be manually specified in the learning algorithm.
- General purpose kernel. It is used when we have no prior knowledge about the data.



How Does SVMs Work?

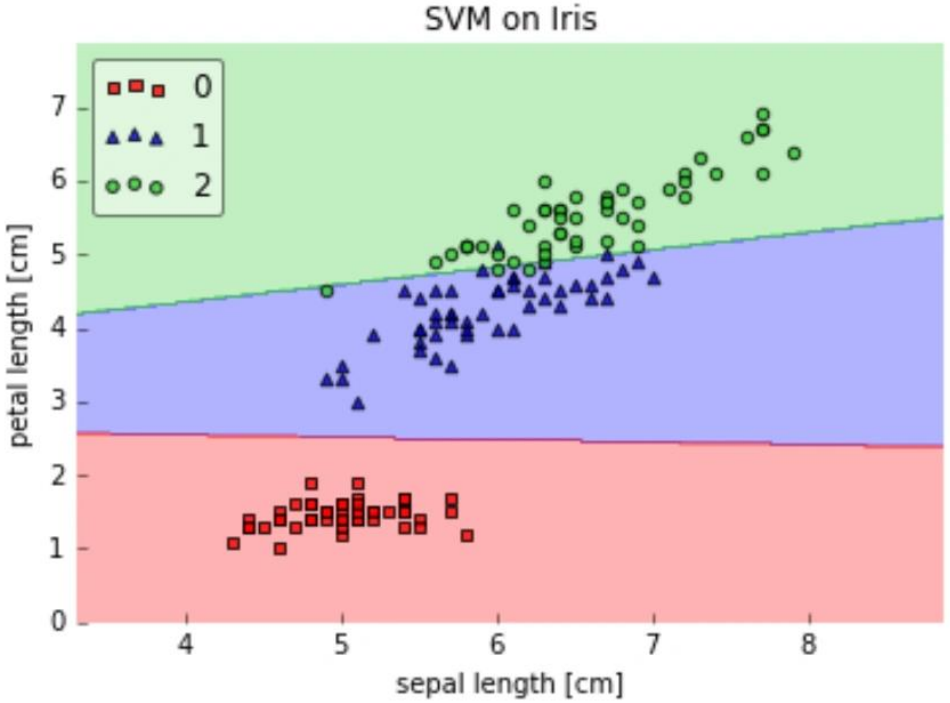
How to Choose the Right Kernel that will work efficiently for the dataset?

It is necessary to choose a good kernel function because the performance of the model depends on it.

Here are the Points to choose the right Kernal:

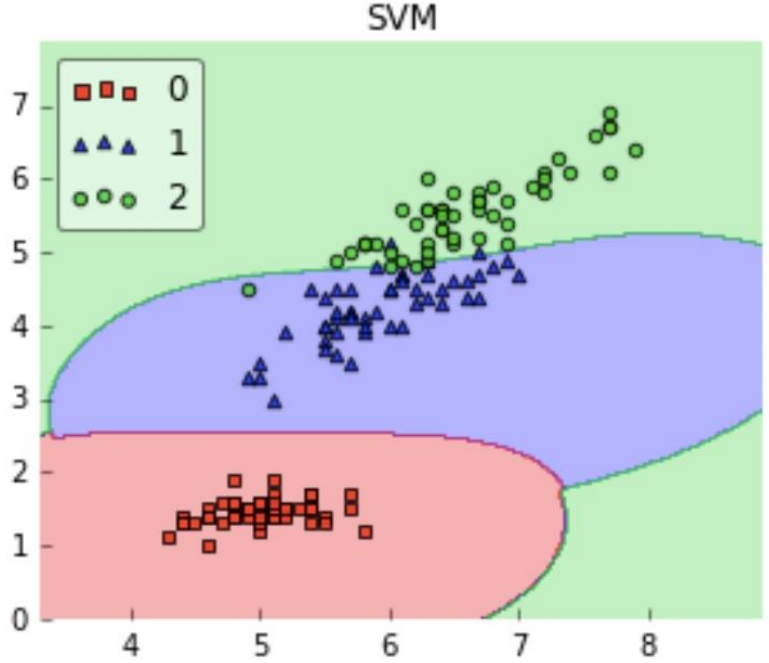
- **Kernel selection depends on the dataset type.**
- **For linearly separable data, use a linear kernel:**
  - It is simple and has lower complexity compared to other kernels.
  - Start by assuming your data is linearly separable and try the linear kernel first.
- **Move to more complex kernels if needed.**
- **Commonly used kernels:**
  - Linear and RBF (Radial Basis Function) are widely used.
  - Polynomial kernels are rarely used due to poor efficiency.
- **If both linear and RBF kernels give similar results:**
  - Choose the simpler option, which is the linear kernel.

How Does SVMs Work? – Example 1



- Only take 2 features that mean 2 dimensions only. Figure plotted the decision boundary of a linear SVM on 2 features of the iris dataset.
- Here we see that a linear kernel works fine on this dataset, but now let's see how will RBF kernel work.

How Does SVMs Work? – Example 1



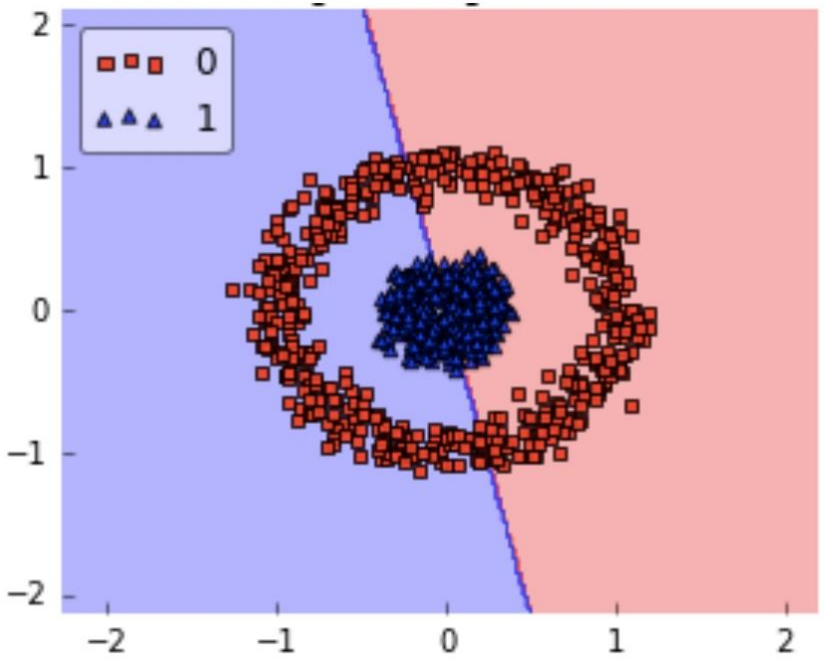
- We can observe that both the kernels give similar results, both work well with our dataset but which one should we choose?
- Linear SVM is a parametric model.
- A Parametric Model is a **concept used to describe a model in which all its data is represented within its parameters**.
- In short, the only **information needed to predict the future** from the current value is the **parameters**.

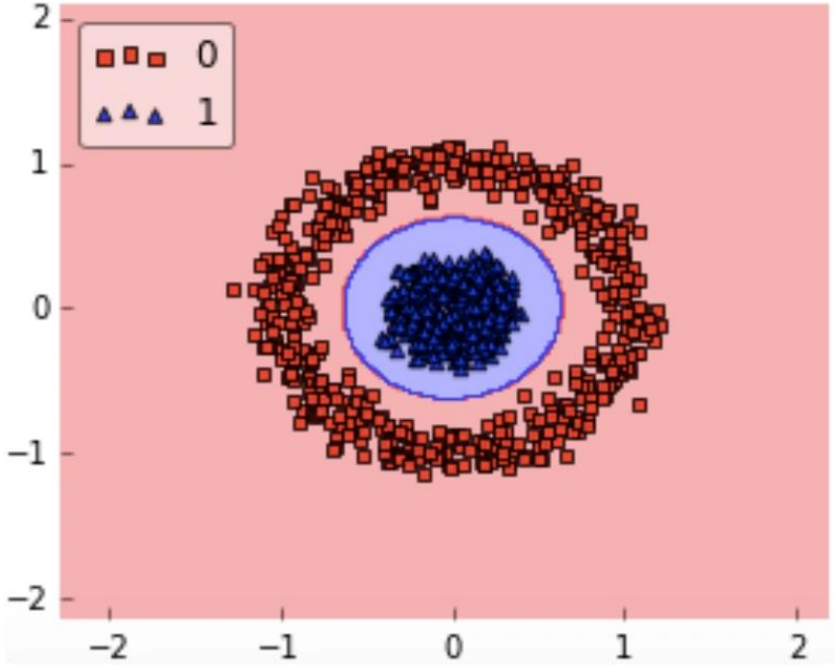
How Does SVMs Work?

Here are some points you should go through:

- The complexity of the RBF kernel increases with the size of the training data.
- Preparing the RBF kernel is computationally expensive.
- The kernel matrix must be stored and maintained, which requires additional memory.
- Projection into the “infinite” higher-dimensional space (where data becomes linearly separable) is costly, especially during prediction.
- Using a linear kernel on a non-linear dataset results in very low accuracy and is not suitable.

How Does SVMs Work? – Example 2





- So, for this kind of dataset, we can use RBF without even a second thought because it makes decision boundary like this.

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Advantages

1. **High-Dimensional Performance:** Performs well even when there are more features (dimensions) than samples, making it useful for tasks like text classification or image recognition. Also, suitable for image classification and gene expression analysis.

1. **Nonlinear Capability:** Great for datasets where the separation between categories is not clear. It can handle both linear and non-linear data effectively.

1. **Outlier Resilience:** The soft margin feature allows SVM to ignore outliers, enhancing robustness in spam detection and anomaly detection.

1. **Binary and Multiclass Support:** SVM is effective for both binary classification and multiclass classification, suitable for applications in text classification.

Advantages

4. **Memory Efficiency:** Focuses on support vectors, making it memory efficient compared to other algorithms.

4. **Avoids overfitting**: Focuses on finding the best decision boundary (margin) between classes, which helps in reducing the risk of overfitting, especially in high-dimensional data.

4. **Versatile with kernels**: By using different kernel functions, SVM can adapt to various types of data and solve complex problems. Different kernels can be used for decision functions; it can be specified or customized.

4. **Robust to outliers:** SVM is less affected by outliers because it focuses on the support vectors (data points closest to the margin), which helps in creating a more generalized model.

Topic 9: Supervised Learning Modelling And Evaluation: Classification Using Support Vector Machine

- Introduction to SVM
- Terminology
- Application
- Types
- How does SVMs work?
- Advantages
- Disadvantages

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Disadvantages

1. **Slow Training**: SVM can be computationally expensive and slow to train, especially when the dataset is very large.

1. **Parameter Tuning Difficulty**: Selecting the right **kernel** and adjusting parameters like **C** requires careful tuning, which often requires a lot of trial and error, impacting **SVM algorithms.**

1. **Noise Sensitivity**: If the dataset has too many overlapping classes or noise, SVM may struggle to perform well because it tries to find a perfect separation.

1. **Limited Interpretability**: The complexity of the **hyperplane** in higher dimensions makes SVM less interpretable than other models.

1. **Feature Scaling Sensitivity**: Proper **feature scaling** is essential; otherwise, SVM models may perform poorly.

THANK YOU



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